

OmniFed: Towards Configurable Cross-Silo Federated Learning

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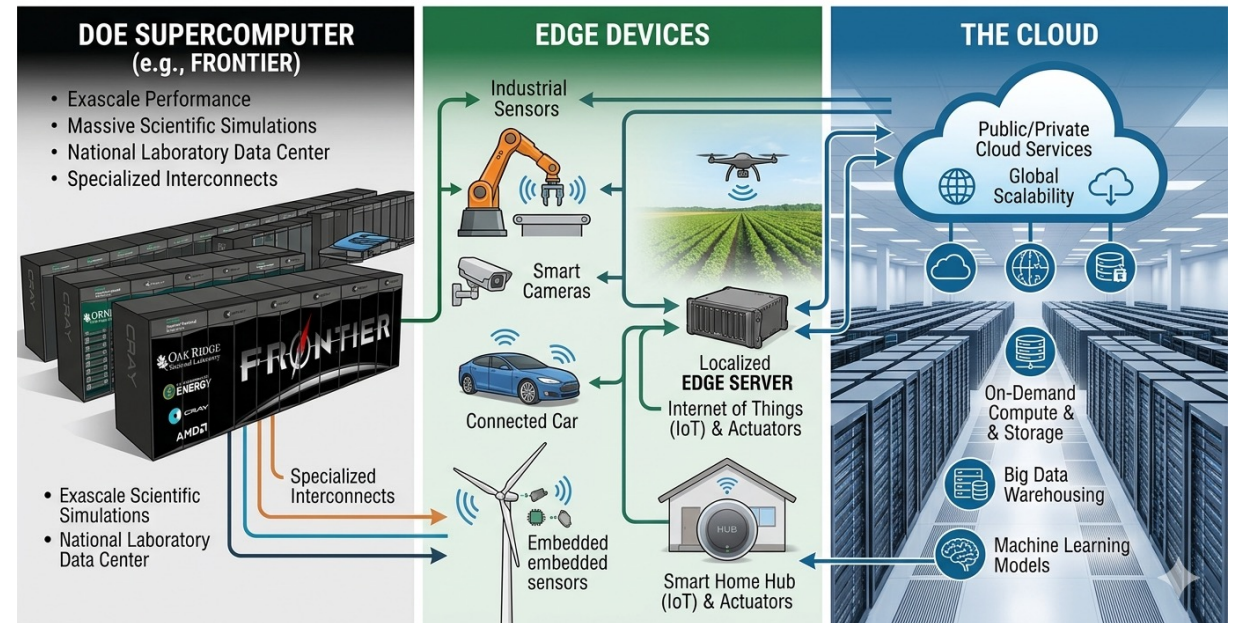
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Trajectory of Federated Learning

- SoTA Models already pre-trained on the internet-scale data; next frontier lies behind silos
- Federated learning (FL) is more than mobile/edge devices training with a single server
- Expansion of the compute continuum; current research environment itself is more heterogeneous than ever!

THE COMPUTING CONTINUUM: COMPARING DOE SUPERCOMPUTING, EDGE, AND CLOUD



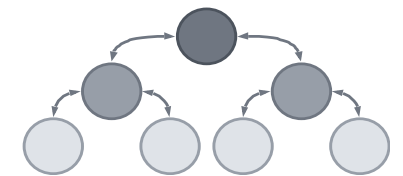
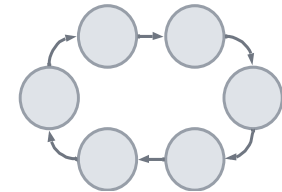
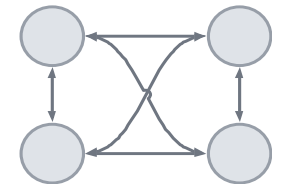
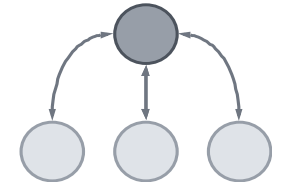
Challenges



- Opportunities come with challenges
 - Data heterogeneity
 - Compute heterogeneity
 - Network heterogeneity
 - Privacy, security and authentication
 - Scalability
- To meet these challenges, tools/platforms need to make explicit design trade-offs:
 - production readiness vs. flexibility
 - Ease-of-use vs. configurability
 - Simulation vs. deployment

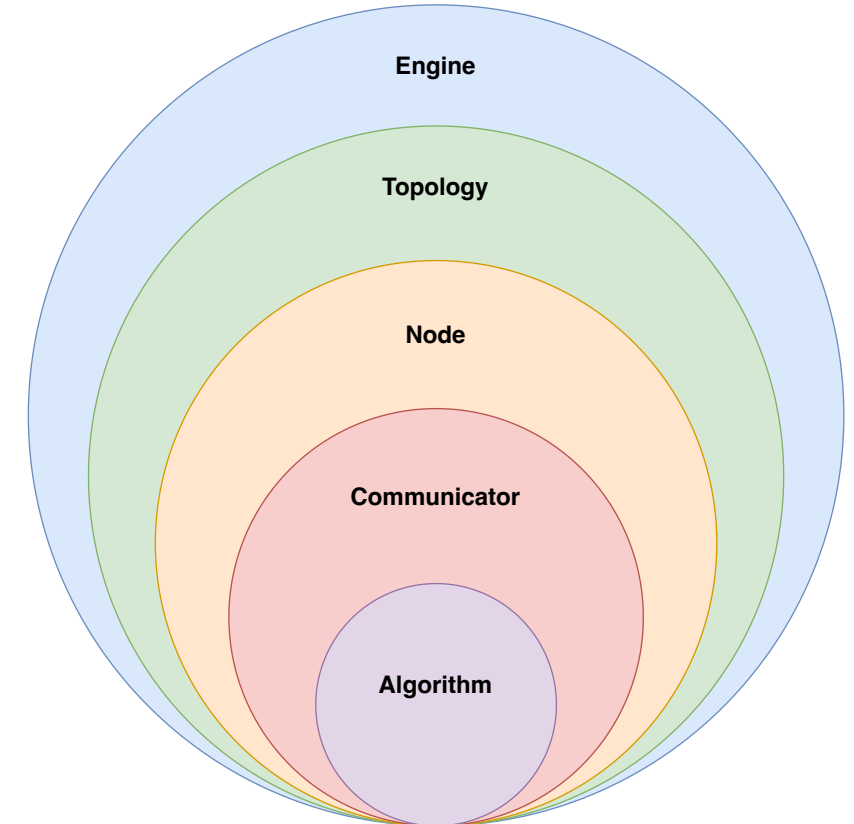
Motivations

- Rapid FL prototyping/testing with minimal boilerplate
- Flexibility for complex topologies
 - Centralized, decentralized, hierarchical
- Configuration-driven, schema-based deployment for reproducibility
 - Easily add/remove features like privacy, compression
- Swap-out design with clear separation of concerns
 - Separate compute resource, communication backend, algorithm strategy



Abstractions for configurable and flexible FL

- Core abstractions:
 - Algorithm
 - Communicator
 - Node
 - Topology
 - Engine
- Clear separation of concerns with minimal interfaces
- Override-what-you-need paradigm



Design Motivation Example

- Deploying FL algorithms with single file algorithm plugins

FedAvg

```
defaults:
- override topology: centralized
- override algorithm: fedavg
- override model: resnet152
- override datamodule: cifar100

topology:
_target_: src.omnifed.topology.CentralizedTopology
num_clients: 8
inner_comm:
_target_: src.omnifed.communicator.GrpcCommunicator
master_port: 50051
master_addr: 127.0.0.1

algorithm:
_target_: src.omnifed.algorithm.FedAvg
lr: 0.01

global_rounds: 2
```

Scaffold

```
defaults:
- override topology: centralized
- override algorithm: fedavg
- override model: resnet152
- override datamodule: cifar100

topology:
_target_: src.omnifed.topology.CentralizedTopology
num_clients: 8
inner_comm:
_target_: src.omnifed.communicator.GrpcCommunicator
master_port: 50051
master_addr: 127.0.0.1

algorithm:
_target_: src.omnifed.algorithm.Scaffold
lr: 0.01

global_rounds: 2
```

Design Motivation Example cont'd...

- Testing compression w.r.t. communication savings vs. convergence quality

```
inner_comm:
  _target_: src.omnifed.communicator.TorchDistCommunicator
  port: 28670
  compression:
    _target_: src.omnifed.communicator.compression.TopK
    k: 0.001
```

Sparsification

Quantization

```
inner_comm:
  _target_: src.omnifed.communicator.GrpcCommunicator
  port: 28670
  compression:
    _target_: src.omnifed.communicator.compression.QSGD
    bit_width: 8
```

```
inner_comm:
  _target_: src.omnifed.communicator.TorchDistCommunicator
  port: 28670
  compression:
    _target_: src.omnifed.communicator.compression.PowerSGD
    rank: 16
```

Low-rank approximation

Design Motivation Example



- Evaluating privacy-preserving mechanisms

```
inner_comm:
  _target_: src.omnifed.communicator.TorchDistCommunicator
  port: 28670
  privacy:
    _target_: src.omnifed.privacy.DifferentialPrivacy
    epsilon: 1.0
    delta: 0.00001
```

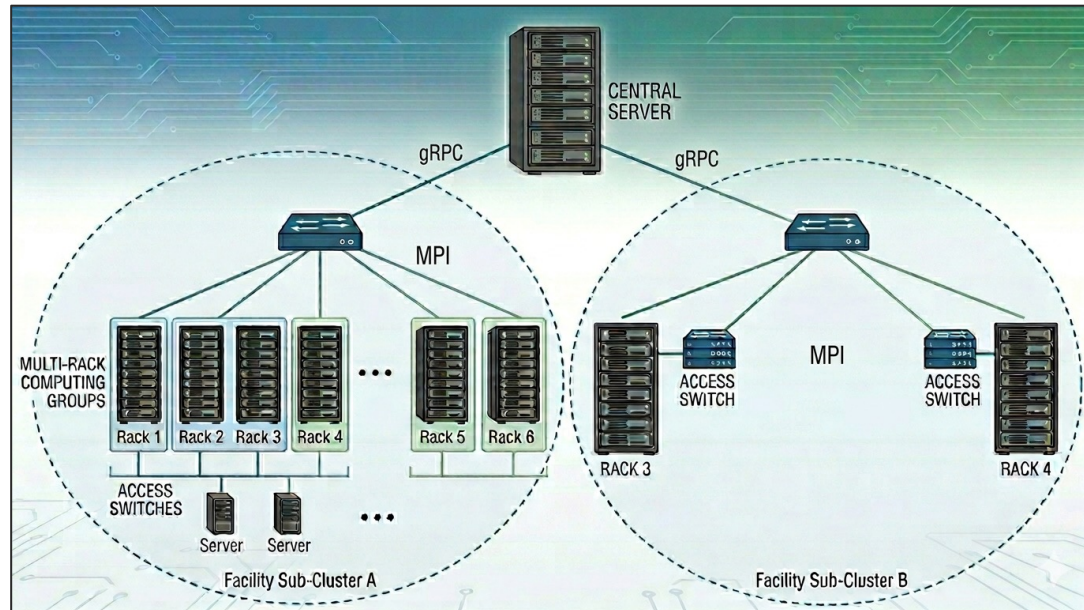
Noise injection-based

```
inner_comm:
  _target_: src.omnifed.communicator.TorchDistCommunicator
  port: 28670
  privacy:
    _target_: src.omnifed.privacy.HomomorphicEncryption
    polymod_degree: 16384
```

Cryptography-based

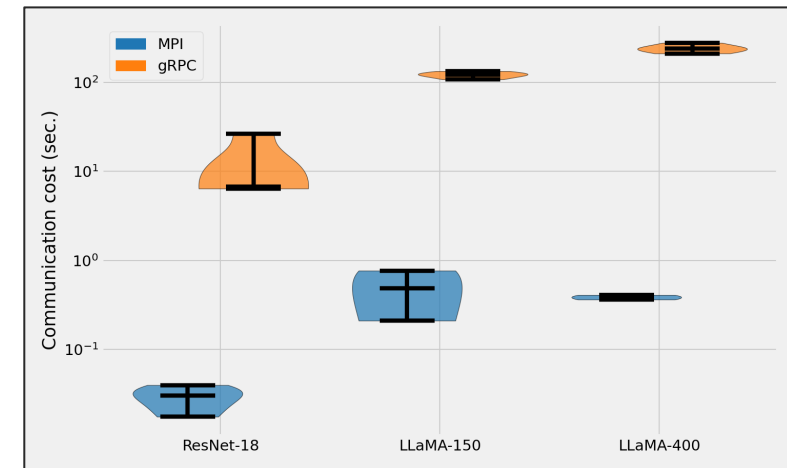
Use-case: Simulating Cross-silo FL

Hierarchical cross-silo training

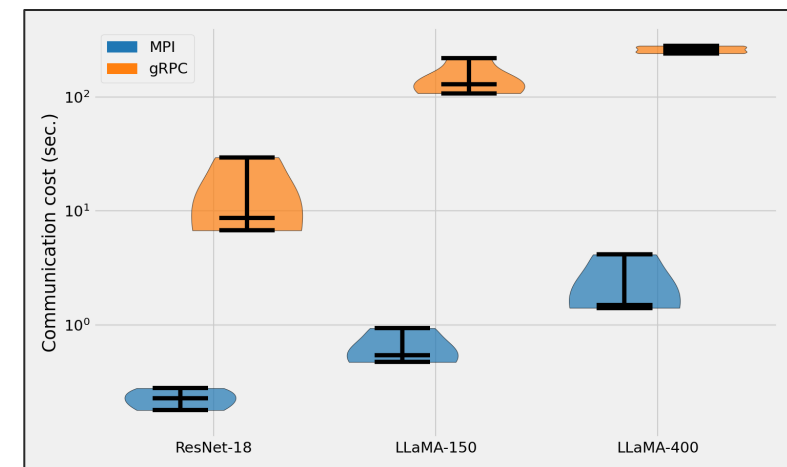


- Model cross-silo training as hierarchical FL
- Mix multiple communication protocol
- Tested on 16/128 nodes on Frontier @OLCF

16 Frontier nodes

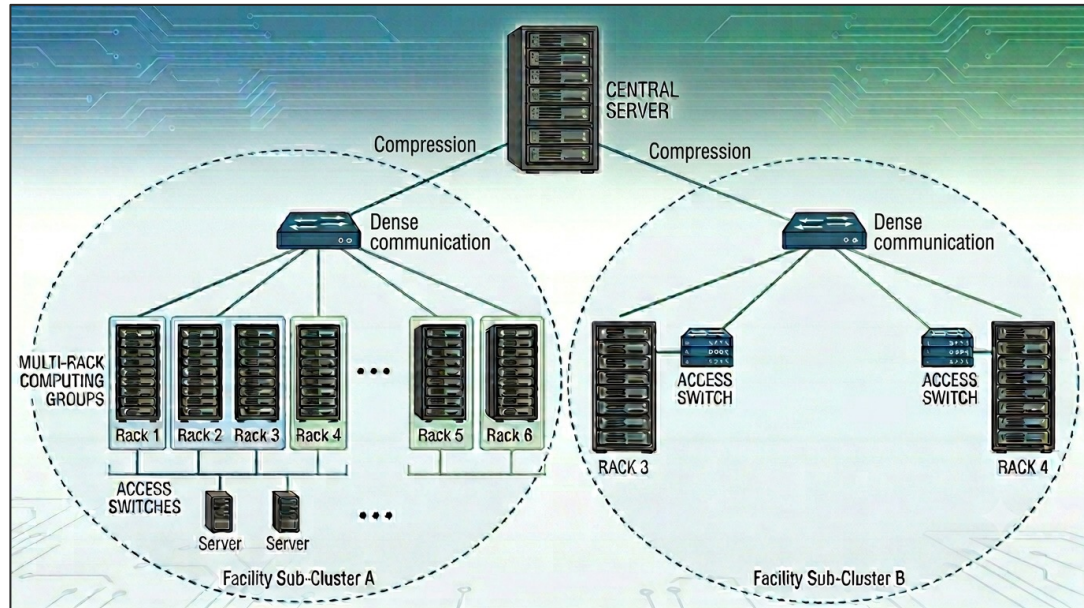


128 Frontier nodes



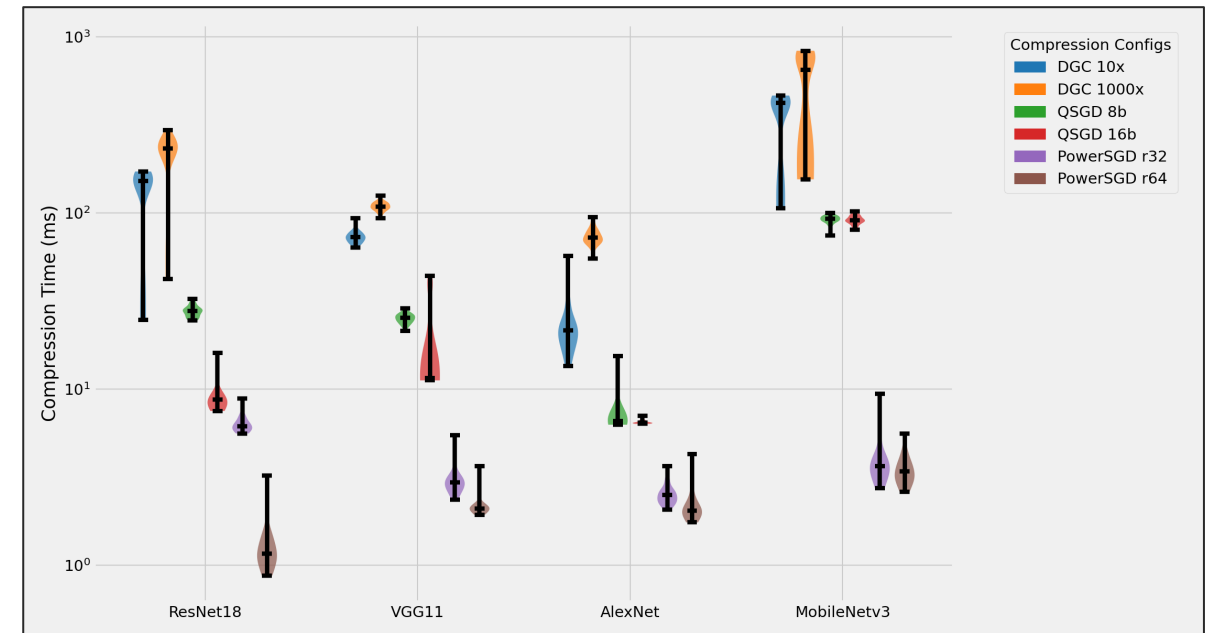
Use-case: Simulating Cross-silo FL cont'd...

Varying compression over sub-aggregators



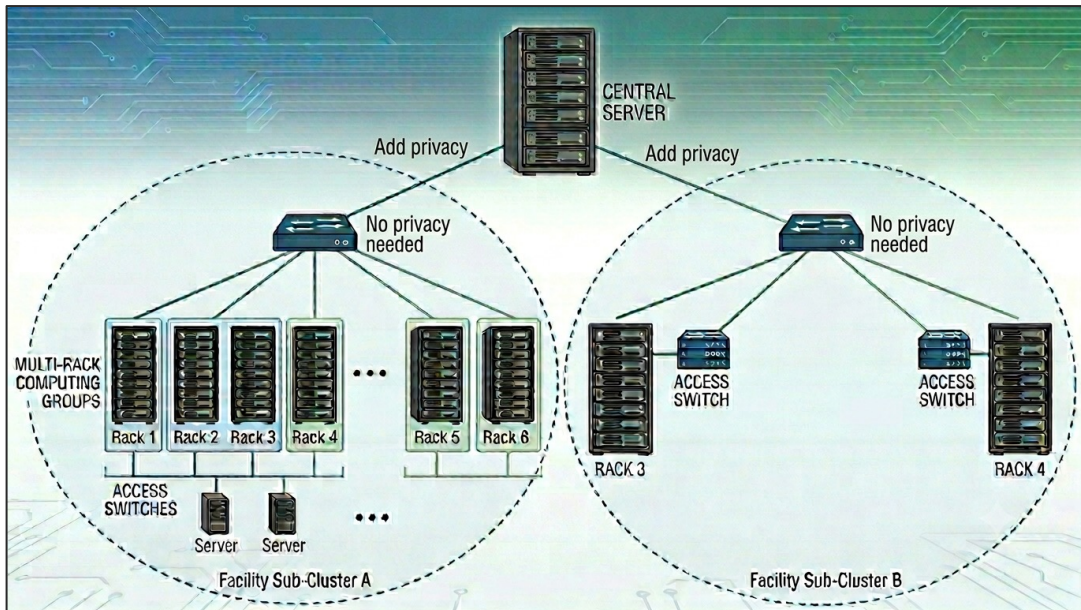
- Adaptive communication across sub-aggregators
- Tested on NVIDIA DGX server

Comparing compression overhead



Use-case: Simulating Cross-silo FL cont'd...

Adding privacy ONLY at infrastructure boundaries



Comparing overhead of privacy mechanisms

Model	Privacy overhead (seconds)	
	Differential Privacy	Homomorphic Encryption
ResNet-18	1.45	68.72
VGG-11	14.4	786
AlexNet	6.9	458.7
MobileNetv3	1.2	29.8

Varying Different Privacy parameters to compare model utility

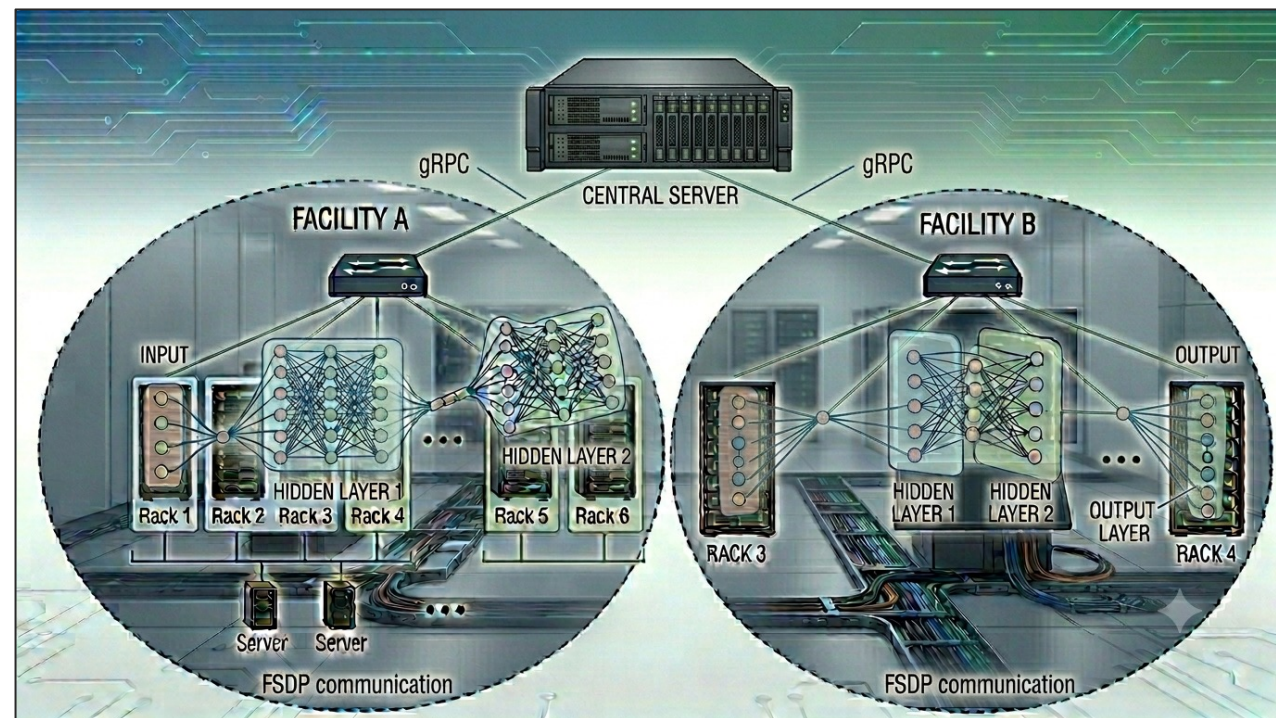
Model	DP Test accuracy	
	$\epsilon=1$	$\epsilon=10$
ResNet-18	97.98%	98.06%
VGG-11	24.1%	28.6%
AlexNet	16.31%	39.54%
MobileNetv3	23.72%	58.8%

- Adaptive privacy mechanisms across silos
- Data secure within a closed system; vulnerable outside

Use-case: Federated AI for LLMs

- Frontier/foundation models fine-tuned on scientific data stored behind secure silos
- Model-size not just a communication, but a computation bottleneck as well!
- Already optimized in HPC AI training!

Training/fine-tuning models too large for individual compute endpoints



Takeaways



- FL is underexplored on SoTA AI over scientific data residing behind closed silos
- Although always relevant, FL is scaling beyond clients training locally, and a central server aggregating globally
- Training configurations may comprise of *many* clients with *fewer* resources, and more recently, *fewer* clients with *larger* resource pools
- Research environment itself is heterogeneous, spanning edge, cloud and leadership-class HPC, and thus, should be *modular, flexible* and *configurable*
- Users/engineers/researchers should be able to *easily* reconfigure the entire FL stack (topology, communication, engine, algorithm, privacy)



Tyagi, S., Cozma, A., Kotevska, O., & Wang, F. (2025).
OmniFed: A Modular Framework for Configurable Federated Learning from Edge to HPC. SC25-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, 516-523.


```

1 topology:
2   # total processes launched
3   world_size: 7
4
5   rpc:
6     server_rank: 0
7     addr: "127.0.0.1"
8     port: 50051
9     client_ranks: [1, 2]  # facility leaders that talk to gRPC server
10
11   facilities:
12     - name: "fac1"
13       mpi:
14         addr: "127.0.0.1"
15         port: 28250
16         world_size: 3
17         members: [1, 3, 4]  # global ranks in this facility
18         leader_rank: 1      # must be in members
19     - name: "fac2"
20       mpi:
21         addr: "127.0.0.1"
22         port: 28290
23         world_size: 3
24         members: [2, 5, 6]
25         leader_rank: 2
26
27   communicators:
28     intra_facility: torch_mpi  # Torch process group inside each facility (declarative).
29     global_aggregation: grpc   # Flora gRPC daemon + facility leaders as clients.
30
31   training:
32     # how many actual workers
33     dataset_total_clients: 6

```